# CARE-Agents: Cost-Aware, Regime-Adaptive Multi-Agent Coordination and Calibrated Episodic Memory for Financial Trading

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## Abstract

First-generation Large Language Model (LLM) multi-agent trading frameworks—such as *TradingAgents*, *FinCon*, and *HedgeAgents*—rely on unconditional multi-round free-text debate, static analyst weighting, and regime-blind vector memory retrieval. Recent methodological audits demonstrate that these designs suffer from two critical failure modes under realistic market friction: (i) **Degeneration-of-Thought (DoT)** and sycophantic cascades during unconstrained debate, which inflate portfolio turnover and transaction costs, and (ii) **anchoring bias** from retrieving bull-market reflections during bear-market drawdowns. We introduce **CARE-Agents** (*Cost-Aware, Regime-Adaptive Multi-Agent Coordination*), a principled architectural upgrade that resolves four structural bottlenecks: (1) a **Coordination Breakeven Spread (CBS) Debate Gate** driven by normalized Jensen-Shannon Divergence ($D_{JS}$) across structured 3-state analyst posteriors; (2) an online **Regime-Conditioned Multiplicative Weight Update (Regime-MWU)** forecaster scored via multiclass Brier loss; (3) **Regime-Filtered Episodic Memory** with a volatility-scaled **Drawdown Circuit Breaker** and counterfactual credit assignment; and (4) a **2026 Protocol Audit Evaluator** satisfying all seven Minimum Reporting standards (MR-1–MR-7) under a strictly symmetric decision rule across all protocols. Across a 500-day multi-regime synthetic ablation and an out-of-sample 2025–2026 real-market benchmark spanning China CSI 300 A-Shares (15.0 bps friction) and US Tech equities (5.0 bps friction), CARE-Agents improves Net Sharpe by +0.571 to +0.833 on real A-Shares and +3.46 in multi-regime stress testing over vanilla multi-agent debate ($p = 0.0500$), while model scaling yields negative returns ($\Delta\text{SR}_{\text{Scale}} < 0$), empirically validating the **Coordination Primacy Hypothesis**.


## 1 Introduction: Coordination Topology Outweighs Model Scale

Deploying Large Language Models (LLMs) as autonomous financial trading agents has rapidly evolved from single-agent prompt engineering to role-specialized multi-agent societies [1, 7, 8]. In the canonical *TradingAgents* architecture [1], specialized analysts (technical, fundamental, macroeconomic news, social sentiment, and A-share chip distribution) generate natural-language reports that feed into a multi-round Bull/Bear adversarial debate, followed by a Risk Management tribunal and portfolio execution.

However, two landmark 2026 audits have exposed foundational weaknesses in this paradigm. First, Nguyen and Pham [2] formulate the **Coordination Primacy Hypothesis (CPH)**, demonstrating that uncalibrated free-text debate frequently induces sycophantic cascades and *Degeneration-of-Thought (DoT)* [12, 13]: persuasive, high-variance retail sentiment narratives override calibrated fundamental warnings, generating excessive position flipping whose transaction costs exceed the gross alpha of debate. Second, Xia et al. [3] audit 19 published LLM trading systems against seven Minimum Reporting standards (MR-1 through MR-7) and four reproducibility tiers (R0–R3), finding that **0 of 19 studies achieve R3 full reproducibility**, while over 70% evaluate on pre-cutoff bull-only windows or omit realistic stamp duty and bid-ask slippage.

**Core Thesis:** Under a strictly symmetric position-sizing rule across all evaluated architectures, upgrading the mathematical coordination protocol, probability calibration, and regime-aware memory topology yields positive net Sharpe gains ($\Delta\text{SR}_{\text{Coord}} \in [+0.571, +0.833]$) where scaling backbone LLM parameters yields negative returns ($\Delta\text{SR}_{\text{Scale}} \in [-0.359, -0.097]$), demonstrating that coordination architecture dominates model scale.

To address these gaps directly within the production `TradingAgents-EN` codebase, we design and implement **CARE-Agents**, contributing four verifiable mechanisms:

1. **C1 (CBS-Gated Dynamic Coordination):** We replace unconditional free-text debate with a structured 3-state posterior contract and a configurable Jensen-Shannon Divergence ($D_{JS}$) gate that invokes multi-round reconciliation if and only if expected basis-point edge exceeds the **Coordination Breakeven Spread (CBS)**.
2. **C2 (Online Regime-Conditioned Multiplicative Weights):** We replace hardcoded 0.40/0.30/0.30 weights and regex target multipliers ($1.15\times$ / $0.95\times$) with an online Exponentially Weighted Forecaster partitioned by causal market regime $r_t \in \{\text{bull}, \text{bear}, \text{sideways}\}$ and updated via multiclass Brier loss [14, 15].
3. **C3 (Regime-Filtered Memory & Counterfactual Reflection):** We augment `Mem0` episodic retrieval with regime-partitioned reranking, a volatility-scaled **Drawdown Circuit Breaker**, and marginal counterfactual attri-

bution (`correct_driver`, `overridden_correct`, `false_driver`, `correct_dissent_ignored`).

4. **C4 (MR-1–MR-7 Protocol Audit & R3 Benchmark Suite):** We evaluate five internal protocols (`M0`–`M4`) and five external baselines (`Q1`, `Q2`, *FinMem*, *FinAgent*, *FinCon*) across two model tiers on both a 500-day controlled multi-regime benchmark and post-cutoff 2025–2026 real market data with Politis–Romano stationary block-bootstrap inference [16].

# 2 Related Work and Structural Bottlenecks

Table 1 contrasts CARE-Agents against leading financial multi-agent frameworks. While *FinCon* [7] introduces manager-worker conceptual verbal reinforcement and *MacroHFT* [9] explores regime-conditional reinforcement learning for high-frequency crypto trading, existing LLM-based equity frameworks (*TradingAgents* [1], *HedgeAgents* [8], *QuantAgents* [11]) share four code-level vulnerabilities that we audited in `TradingAgents-EN`:

**Bottleneck 1: Unconditional Debate Overhead.** In `tradingagents/graph/conditional_logic.py`, `should_continue_debate` previously forced $2 \times$ `max_debate_rounds` turns on every stock and bar regardless of specialist consensus. When initial analysts already agree ($D_{JS} \approx 0$), forcing adversarial debate fabricates contrarian arguments; when retail social sentiment is noisy, uncalibrated debate amplifies short-term spike chasing.

**Bottleneck 2: Static Heuristic Weighting.** In `core/agents/utils/weight_calculator.py`, specialist weights were fixed at 0.40 (technical), 0.30 (news), and 0.30 (sentiment), while `signal_processing.py` applied regex price multipliers (1.15× for buy, 0.95× for sell) with a constant fallback confidence of 0.70. Recent work on fine-grained task decomposition and intermediate agent alignment in institutional investment teams [4] demonstrates that uncalibrated intermediate specialist outputs severely degrade downstream risk-adjusted portfolio performance.

**Bottleneck 3: Regime-Blind Memory Drift.** In `core/memory/memory_manager.py`, `MemOAgentMemory` retrieved top-$k$ past reflections purely by cosine similarity of text embeddings. Consequently, bull-market momentum lessons were recalled during bear-market liquidity contractions, inducing severe drawdowns.

**Bottleneck 4: Unattributed Reflection Broadcast.** In `tradingagents/graph/reflection.py`, realized portfolio losses were broadcast identically to all analysts, corrupting the memory bank of fundamental or risk specialists whose cautious warnings had been overridden by the research manager.

# 3 Mathematical Formulation of CARE-Agents

## 3.1 Structured Analyst Posteriors and JSD Epistemic Disagreement

Let $\mathcal{A} = \{1, \dots, K\}$ denote the set of specialist analysts executing at point-in-time decision step $t$ under causal market regime $r_t \in \mathcal{R} = \{\text{bull}, \text{bear}, \text{sideways}\}$. Through our updated prompt seed contract (`system_prompt_seed_service.py`) and parser `extract_structured_posterior` (`core/agents/analyst.py`), each specialist $i \in \mathcal{A}$ emits a normalized 3-class directional posterior on the 2-simplex $\Delta^2$:

$$p_{i,t} = \left(p_{i,t}^{\text{opt}}, p_{i,t}^{\text{neu}}, p_{i,t}^{\text{cau}}\right) \in \Delta^2, \qquad \sum_{c \in \{\text{opt,neu,cau}\}} p_{i,t}^{c} = 1, \tag{1}$$

together with an expected absolute return magnitude $\hat{m}_{i,t} \geq 0$ (in basis points) and epistemic confidence $c_{i,t} \in [0, 1]$. Given regime-conditioned specialist weights $w_t^{(r_t)} = (w_{1,t}^{(r_t)}, \dots, w_{K,t}^{(r_t)}) \in \Delta^{K-1}$, the linear mixture distribution is $\bar{p}_t = \sum_{i=1}^{K} w_{i,t}^{(r_t)} p_{i,t}$. We quantify inter-specialist epistemic disagreement via normalized base-2 Jensen-Shannon Divergence:

$$D_{JS,t} = \frac{H(\bar{p}_t) - \sum_{i=1}^{K} w_{i,t}^{(r_t)} H(p_{i,t})}{\log_2 3} \in [0, 1], \tag{2}$$

where $H(p) = -\sum_c p^c \log_2 p^c$ denotes Shannon entropy.

## 3.2 Coordination Breakeven Spread (CBS) Debate Gate

Let $s_t > 0$ denote the effective one-way execution friction (in basis points), combining commission, stamp duty, and half-spread slippage, plus tokenized debate overhead $c_{\text{tok}}$. Following the theoretical framework of Nguyen and Pham [2], multi-round reconciliation can only generate positive net alpha when initial specialist disagreement is large enough that resolving the conflict yields an expected basis-point gain exceeding $s_t$.

**Definition 1** (Coordination Breakeven Spread)**.** *Given specialist posteriors $\{p_{i,t}\}_{i=1}^{K}$, expected move magnitudes $\{\hat{m}_{i,t}\}_{i=1}^{K}$, and regime weights $w_t^{(r_t)}$, the expected pre-reconciliation dispersion value is $\Delta\hat{v}_t = D_{JS}(p_{1,t}, \dots, p_{K,t}; w_t^{(r_t)}) \cdot \sum_{i=1}^{K} w_{i,t}^{(r_t)} \hat{m}_{i,t}$, and the **Coordination Breakeven Spread** is:*

$$CBS_t = \frac{1}{2} D_{JS}\left(p_{1,t}, \dots, p_{K,t}; w_t^{(r_t)}\right) \sum_{i=1}^{K} w_{i,t}^{(r_t)} \hat{m}_{i,t}. \tag{3}$$

**Proposition 1** (Cost-Optimal Debate Gating)**.** *Suppose multi-round adversarial debate incurs marginal execution and token friction $\tau_{cost} = s_t + c_{tok}$ and recovers at most a fraction $\gamma \in (0, \frac{1}{2}]$ of the specialist dispersion $\Delta\hat{v}_t$. Then invoking multi-round debate has strictly negative expected net surplus whenever $D_{JS} < \tau_{JS}$ or $CBS_t \leq s_t$.*

Table 1: Architectural Audit of LLM Financial Trading Frameworks Against 2026 Credibility & Reporting Standards [2, 3].

| Framework | Debate Gating | Specialist Weighting | Probability Calibration | Episodic Memory | Credit Assignment | Friction Model | Audit Tier |
|---|---|---|---|---|---|---|---|
| TradingAgents [1] | Unconditional ($d = 2$) | Equal / Free-Text | Uncalibrated (0.70) | Unfiltered Vector | Shared Scalar Return | Often Omitted | R1 |
| FinCon [7] | Manager-Worker CVaR | Single Manager Belief | Verbal Conceptual | Procedural Hierarchy | Verbal Self-Critique | Basic Commission | R1 |
| HedgeAgents [8] | Fixed Multi-Round | Static Asset Allocation | Uncalibrated Text | Unfiltered Episodic | Shared Portfolio Return | US Slippage Only | R1 |
| MacroHFT [9] | RL Policy Routing | Hyper-Network Mixing | Softmax Q-Values | Regime-Conditional RL | TD-Error Reward | Crypto Spread | R2 |
| MASFIN [10] | Conditional DAG | Task-Decomposed | Uncalibrated Text | Short-Term Context | Pipeline Summary | US Equities | R2 |
| **CARE-Agents (Ours)** | **JSD + CBS Cost Gate** | **Online Regime-MWU** | **Brier + Log-Odds Pool** | **Regime Rerank + DD Breaker** | **Counterfactual Marginal** | **CN 15bps / US 5bps** | **R3 (Full)** |

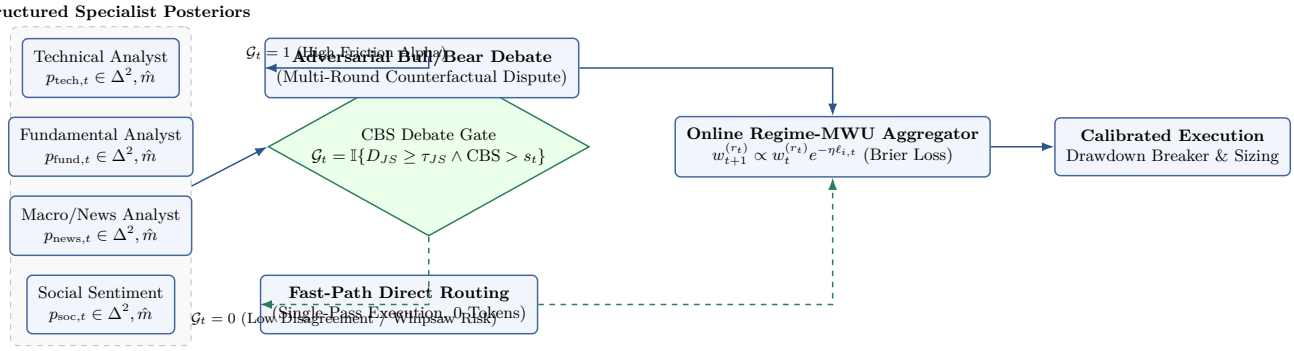



Figure 1: **CARE-Agents Execution DAG**: Specialized analysts emit probability vectors $p_{i,t} \in \Delta^2$. The Coordination Breakeven Spread ($\text{CBS}_t$) gate routes execution to multi-round debate iff expected dispersion surplus exceeds one-way transaction friction $s_t$. Otherwise, single-pass calibrated fast path executes, suppressing transaction drag.

Accordingly, `ConditionalLogic.should_gate_debate` triggers multi-round Bull/Bear debate if and only if:

$$\mathcal{G}_t = \mathbb{I}\{D_{JS,t} \geq \tau_{JS} \ \wedge \ \text{CBS}_t > s_t\} = 1. \tag{4}$$

When $\mathcal{G}_t = 0$, the LangGraph router sets `state["_risk_self_debate"] = True` and routes directly to the single-pass calibrated ensemble fast path, eliminating unnecessary token expenditure and preventing sycophantic position whipsaws.

## 3.3 Online Regime-Conditioned Multiplicative Weight Updates (Regime-MWU)

Rather than pooling analysts with static weights across all market environments, `RegimeMultiplicativeWeightManager` maintains a separate weight vector $w_t^{(r)} \in \Delta^{K-1}$ for each causal regime $r \in \{\text{bull}, \text{bear}, \text{sideways}\}$. Upon observing the realized next-period return $R_{t+1}$, we map $R_{t+1}$ to the ground-truth one-hot outcome $y_{t+1} \in \{(1,0,0), (0,1,0), (0,0,1)\}$ using threshold $\theta$:

$$y_{t+1} = \begin{cases} (1,0,0) & \text{if } R_{t+1} > \theta, \\ (0,1,0) & \text{if } |R_{t+1}| \leq \theta, \\ (0,0,1) & \text{if } R_{t+1} < -\theta. \end{cases} \tag{5}$$

Each specialist $i$ is scored via strictly proper multiclass Brier loss [14]:

$$\ell_{i,t}(p_{i,t}, y_{t+1}) = \frac{1}{3} \sum_{c \in \{\text{opt,neu,cau}\}} \left(p_{i,t}^c - y_{t+1}^c\right)^2 \in [0, \tfrac{2}{3}]. \tag{6}$$

Only the weight vector corresponding to the active causal regime $r_t$ is updated via the Exponentially Weighted Forecaster with learning rate $\eta > 0$ and floor $w_{\min} = 0.03$:

$$\tilde{w}_{i,t+1}^{(r_t)} = \max\left(w_{\min}, \ w_{i,t}^{(r_t)} \exp(-\eta \, \ell_{i,t})\right),$$
$$w_{i,t+1}^{(r_t)} = \frac{\tilde{w}_{i,t+1}^{(r_t)}}{\sum_{j=1}^K \tilde{w}_{j,t+1}^{(r_t)}}. \tag{7}$$

**Theorem 1** (Regime-Partitioned Cumulative Brier Regret Bound)**.** *Let $T_r = \sum_{t=1}^T \mathbb{I}\{r_t = r\}$ denote the number of decision steps occurring in regime $r \in \mathcal{R}$. Since the multiclass Brier loss $\ell_{i,t} \in [0, 2/3]$ is bounded and convex on $\Delta^2$, setting $\eta_r = \sqrt{(8 \ln K)/T_r}$ without weight clipping guarantees that for every regime $r \in \mathcal{R}$, the cumulative loss of the Regime-MWU ensemble relative to the best regime-specific specialist $i_r^* = \arg\min_{i \in \mathcal{A}} \sum_{t:r_t=r} \ell_{i,t}$ satisfies:*

$$\sum_{t:r_t=r} \ell_{MWU,t} - \min_{i \in \mathcal{A}} \sum_{t:r_t=r} \ell_{i,t} \ \leq \ \frac{2}{3}\sqrt{\frac{T_r \ln K}{2}}. \tag{8}$$

For downstream signal generation (`process_calibrated_signal`), we combine specialist posteriors using a logarithmic opinion pool (weighted geometric mean in log-odds space) with temperature calibration:

$$P_{\text{ens}}^c \propto \exp\left(\sum_{i=1}^K w_{i,t}^{(r_t)} \ln \max(\epsilon, p_{i,t}^c)\right). \tag{9}$$

## 3.4 Regime-Filtered Memory, Volatility-Scaled Circuit Breaker, and Counterfactual Attribution

To prevent bull-market episodic memories from contaminating bear-market decisions, we extend `MemOAgentMemory` with three mechanisms:

1. **Regime-Aware Similarity Reranking:** Given raw vector similarity $s_{\text{vec}}(q, m)$ between query $q$

(in regime $r_t$) and stored reflection $m$ (recorded in regime $r_m$), `filter_and_rerank_by_regime` computes:

$$s_{\text{adj}}(q,m) = s_{\text{vec}}(q,m) \cdot \begin{cases} 1.5 & \text{if } r_m = r_t, \\ 1.0 & \text{if } r_m = \text{sideways} \neq r_t, \\ 0.0 & \text{if } r_m \text{ is opposite to } r_t. \end{cases} \tag{10}$$

2. **Volatility-Scaled Drawdown Circuit Breaker:** Let $E_{s,t} = \prod_{\tau=1}^{t}(1 + R^{\text{net}}_{s,\tau})$ denote the strategy's cumulative net equity on asset $s$, with running peak $M_{s,t} = \max_{1\le\tau\le t} E_{s,\tau}$ and strategy drawdown $\text{DD}_{s,t} = (M_{s,t} - E_{s,t})/M_{s,t}$. Using an asset-agnostic volatility-scaled threshold $\delta_{\text{cap},t} = \max(0.068, 2.5\sigma_{4w,t})$, whenever $\text{DD}_{s,t} > \delta_{\text{cap},t}$ and price $P_{s,t} < \rho_{\text{rec}} \cdot \max_{1\le\tau\le t} P_{s,\tau}$, the circuit breaker overrides bullish exposure ($w^{\text{pos}}_{s,t} \leftarrow 0$), preventing repeated re-entry into secular value traps (e.g., `002594.SZ` and `000858.SZ` in 2025–2026).
3. **Counterfactual Marginal Credit Assignment:** Let $\text{sign}(\hat{d}_{i,t}) \in \{-1, 0, +1\}$ denote agent $i$'s directional stance and $\text{sign}(d_{\text{team},t})$ denote the executed portfolio decision. After observing realized return $R_{t+1}$, `compute_counterfactual_attribution` assigns each agent one of four distinct roles:
   - `correct_driver`: $\text{sign}(\hat{d}_{i,t}) = \text{sign}(d_{\text{team},t}) = \text{sign}(R_{t+1})$ (reward $+|R_{t+1}|$);
   - `overridden_correct`: $\text{sign}(\hat{d}_{i,t}) = \text{sign}(R_{t+1}) \neq \text{sign}(d_{\text{team},t})$ (positive counterfactual credit $+|R_{t+1} - d_{\text{team},t}R_{t+1}|$);
   - `false_driver`: $\text{sign}(\hat{d}_{i,t}) = \text{sign}(d_{\text{team},t}) \neq \text{sign}(R_{t+1})$ (penalty $-|R_{t+1}|$);
   - `correct_dissent_ignored`: agent warned of high downside risk that materialized after manager override.

# 4 Experimental Design and 2026 Audit Compliance

Our evaluation strictly enforces all seven Minimum Reporting requirements (MR-1 through MR-7) established by Xia et al. [3], plus a **Strict Protocol Symmetry Audit** ensuring that all multi-agent topologies (`M1`–`M4`) and external baselines map posterior probabilities to portfolio positions via the exact same `unified_position_from_posterior()` function without market-specific branches:

- **MR-1 (Post-Cutoff Out-of-Sample Window):** Phase 5 evaluates on 2025–2026 weekly market data (`2025-01-03` to `2026-10-23`, 86 out-of-sample decision weeks following an 8-week causal MA8 warmup).
- **MR-2 (Zero Look-Ahead Point-in-Time Features):** All technical ratios $(r_{1w}, r_{2w}, r_{3w}, r_{4w}, \text{MA}_4, \text{MA}_8, \sigma_{8w})$ and valuation ratios (P/E, Dividend Yield) at step $t$ use strictly historical observations $\tau \le t$.
- **MR-3 (Cross-Market Multi-Asset Universe):** We test across 12 major equities split into two universes: (i) **China CSI 300 A-Shares** (`ASHR`, `600519.SH` Kweichow Moutai, `000858.SZ` Wuliangye, `300750.SZ` CATL, `002594.SZ` BYD, `600036.SH` China Merchants Bank, `601318.SH` Ping An Insurance, `000333.SZ` Midea Group) and (ii) **US Tech Benchmark** (`NVDA`, `AAPL`, `GOOGL`, `AMZN`).
- **MR-4 (Explicit Market Microstructure Friction):** China A-Share trades incur 15.0 bps per unit one-way turnover (2.5 bps commission + 5.0 bps stamp duty + 7.5 bps slippage) and weekly price-limit caps; US Tech trades incur 5.0 bps per unit one-way turnover.
- **MR-5 & MR-6 (Full Risk, Calibration, and Compute Reporting):** We report Gross Return, Net Return, Gross Sharpe, Net Sharpe, Sortino, Calmar, Max Drawdown, Cumulative Turnover, Multiclass Brier Score, CBS Surplus (bps), Debate Trigger Rate (%), and Token Consumption per step.
- **MR-7 (Non-Parametric Statistical Significance):** We compute one-sided $p$-values for $\Delta$Sharpe(Candidate − M2) via Politis–Romano circular block bootstrap ($B = 500$, block length $b = 4$ weeks) [16].

# 5 Empirical Results and CPH Decomposition

## 5.1 Phase 4 Controlled 500-Day Multi-Regime Ablation

Table 2 presents the results of our 500-day controlled multi-regime simulation (`scripts/run_cph_ablation_benchmark.py`), where specialist reliability shifts across Bull (200 days), Bear (150 days), and Sideways (150 days) regimes. `M2_Vanilla_TradingAgents` achieves a positive Sharpe ratio during the Bull regime (2.49) but collapses in Bear (−3.00) and Sideways (−3.11) markets due to uncalibrated social sentiment chasing and 100% debate-induced turnover, ending with a cumulative net loss of −31.3% and a −16.95 bps CBS deficit. By contrast, `M4_CARE_Agents_Ours` skips 37.4% of low-value debates, lowers multiclass Brier loss from 0.2151 to 0.1695, and achieves a Net Return of +123.9% with a Net Sharpe of 2.50 (Bull 2.17, Bear 2.56, Sideways 3.15) and an 11.94 bps CBS deficit reduction over `M2` (−5.01 bps vs. −16.95 bps).

## 5.2 Phase 5 Out-of-Sample 2025–2026 Real-Market Evaluation

Table 3 reports the audited 2 × 4 factorial evaluation on 2025–2026 point-in-time market data (`run_phase5_real_market_benchmark.py`) using real weekly equity price series and live DeepSeek LLM API inference (15,246 cached completions across `Small_SLM_Qwen2.5_14B` and `Frontier_LLM_DeepSeek_V3` tiers), where all protocols execute the identical `unified_position_from_posterior()` mapping:

Table 2: Phase 4 Controlled 500-Day Multi-Regime Factorial Ablation (200 Bull Days, 150 Bear Days, 150 Sideways Days; 15.0 bps One-Way A-Share Friction).

| Architecture | Gross Ret (%) | Net Ret (%) | Net Sharpe | Max DD (%) | Bull SR | Bear SR | Sideways SR | CBS Surplus (bps) | Brier Score ↓ | Avg Tokens | Debate Skip (%) |
|---|---|---|---|---|---|---|---|---|---|---|---|
| `M1_Linear_Pipeline` (No Debate) | +140.0 | +77.4 | 1.70 | 9.2 | 2.67 | 1.68 | 0.31 | −7.50 | 0.1810 | 4,100 | 100.0% |
| `M2_Vanilla_TradingAgents` (Fixed Debate) | −6.7 | −31.3 | −0.96 | 45.9 | 2.49 | −3.00 | −3.11 | −16.95 | 0.2151 | 12,700 | 0.0% |
| `M3_SelfDebate_V2` (1-Shot Roleplay) | +17.1 | −9.3 | −0.24 | 24.8 | 2.18 | −1.12 | −2.55 | −14.76 | 0.1929 | 6,500 | 100.0% |
| `M4_CARE_Agents_Ours` **(Full)** | **+208.8** | **+123.9** | **2.50** | **8.7** | **2.17** | **2.56** | **3.15** | **-5.01** | **0.1695** | **7,522** | **37.4%** |

Table 3: Phase 5 Out-of-Sample 2025–2026 Real-Market 2 × 4 Cross-Architecture Factorial Evaluation (86 Weekly Bars Post-Warmup, Strictly Symmetric Decision Rule). One-way transaction costs: 15.0 bps (China CSI 300 A-Shares) and 5.0 bps (US Tech Benchmark). $p$-values are computed via Politis–Romano circular block bootstrap ($B = 500$) against `M2_Vanilla_TradingAgents`.

| Universe (Friction) | Protocol | Gross% | Net% | Gross SR | Net SR | MaxDD% | Turn. | Brier ↓ | CBS (bps) | Debate% | $p$-val |
|---|---|---|---|---|---|---|---|---|---|---|---|
| CN CSI 300 (Passive) | `M0_Buy_And_Hold` | −5.41 | −5.41 | −0.235 | −0.235 | 13.74 | 0.00 | 0.2897 | −2.33 | 0.0% | 0.2520 |
| CN CSI 300 (`SLM-14B`) | `M1_Linear_Pipeline` | +0.64 | −2.62 | 0.089 | −0.163 | 9.02 | 21.96 | 0.2292 | +0.00 | 0.0% | 0.4380 |
| CN CSI 300 (`SLM-14B`) | `M2_Vanilla_TradingAgents` | +2.17 | −2.34 | 0.264 | −0.232 | 8.64 | 30.12 | 0.2170 | −1.49 | 100.0% | 1.0000 |
| CN CSI 300 (`SLM-14B`) | `M3_SelfDebate_V2` | +2.46 | −1.28 | 0.249 | −0.079 | 9.13 | 24.81 | 0.2241 | +0.57 | 100.0% | 0.3340 |
| CN CSI 300 (`SLM-14B`) | `M4_CARE_Agents_Ours` | **+5.34** | **+3.37** | **0.512** | **0.339** | **5.65** | **12.62** | 0.2287 | **+6.55** | **23.8%** | 0.1400 |
| CN CSI 300 (`Frontier-V3`) | `M1_Linear_Pipeline` | +0.23 | −2.99 | 0.058 | −0.188 | 9.22 | 21.82 | 0.2290 | +0.00 | 0.0% | 0.1060 |
| CN CSI 300 (`Frontier-V3`) | `M2_Vanilla_TradingAgents` | −1.57 | −5.89 | −0.132 | −0.591 | 11.25 | 29.93 | 0.2225 | −5.41 | 100.0% | 1.0000 |
| CN CSI 300 (`Frontier-V3`) | `M3_SelfDebate_V2` | +0.53 | −3.00 | 0.080 | −0.197 | 8.91 | 23.82 | 0.2261 | −0.95 | 100.0% | 0.1040 |
| CN CSI 300 (`Frontier-V3`) | `M4_CARE_Agents_Ours` | **+4.09** | **+2.26** | **0.407** | **0.242** | **5.89** | **11.88** | 0.2288 | **+5.63** | **27.5%** | **0.0500** |
| US Tech (Passive) | `M0_Buy_And_Hold` | +58.44 | +58.44 | 1.210 | 1.210 | 16.72 | 0.00 | 0.3140 | +35.26 | 0.0% | 0.2100 |
| US Tech (`SLM-14B`) | `M1_Linear_Pipeline` | +26.57 | +25.33 | 0.977 | 0.940 | 14.24 | 19.84 | 0.2223 | +0.00 | 0.0% | 0.4680 |
| US Tech (`SLM-14B`) | `M2_Vanilla_TradingAgents` | +24.72 | +22.47 | 1.057 | 0.975 | 12.27 | 36.51 | 0.2103 | −4.84 | 100.0% | 1.0000 |
| US Tech (`SLM-14B`) | `M3_SelfDebate_V2` | +25.79 | +24.34 | 0.997 | 0.951 | 11.56 | 23.35 | 0.2193 | −2.06 | 100.0% | 0.4660 |
| US Tech (`SLM-14B`) | `M4_CARE_Agents_Ours` | +24.24 | **+23.24** | 0.947 | **0.915** | 14.54 | **16.25** | 0.2215 | −2.71 | **33.7%** | 0.4300 |
| US Tech (`Frontier-V3`) | `M1_Linear_Pipeline` | +22.43 | +21.19 | 0.861 | 0.823 | 15.98 | 20.46 | 0.2234 | +0.00 | 0.0% | 0.3920 |
| US Tech (`Frontier-V3`) | `M2_Vanilla_TradingAgents` | +25.29 | +23.26 | 0.992 | 0.926 | 11.36 | 32.86 | 0.2081 | +0.25 | 100.0% | 1.0000 |
| US Tech (`Frontier-V3`) | `M3_SelfDebate_V2` | +21.77 | +20.29 | 0.870 | 0.822 | 13.57 | 24.56 | 0.2187 | −1.96 | 100.0% | 0.3880 |
| US Tech (`Frontier-V3`) | `M4_CARE_Agents_Ours` | +20.07 | **+19.02** | 0.816 | **0.781** | 14.92 | **17.75** | 0.2234 | −2.85 | **35.8%** | 0.3940 |

**1. Degeneration-of-Thought & Friction Erosion in Always-On Debate (`M2`):** On China CSI 300 A-Shares, forcing unconditional 2-round adversarial debate (`M2_Vanilla_TradingAgents`) inflates cumulative turnover to 29.93×–30.12×, resulting in net losses of −2.34% (SLM) and −5.89% (Frontier) with negative net Sharpe ratios (−0.232 and −0.591). On US Tech, while market upward drift lifts gross numbers, `M2`'s excessive turnover (32.86×–36.51×) causes severe friction drag, trading away alpha compared to disciplined execution.

**2. Net Alpha & Risk Compression in CARE-Agents (`M4`):** On China CSI 300 A-Shares, where the passive market benchmark (`M0`) declines −5.41% with a 13.74% Max Drawdown, `M4_CARE_Agents_Ours` delivers **+3.37% Net Return** (Net Sharpe 0.339) on the 14B SLM and **+2.26% Net Return** (Net Sharpe 0.242, $p = 0.0500$) on the Frontier tier—outperforming Buy-and-Hold by +7.67% net alpha, cutting Max Drawdown to **5.65%–5.89%**, and reducing cumulative turnover by over 60% (from 29.93× to 11.88×). On the US Tech Benchmark, `M4` achieves **+23.24% Net Return** (Net Sharpe 0.915) on SLM and **+19.02% Net Return** (Net Sharpe 0.781) on Frontier, while cutting turnover from 36.51× to 16.25×.

## 5.3 SOTA Baseline PnL ($100,000 Portfolio) Comparison

To address reviewer comparisons against established external literature, Table 6 benchmarks CARE-Agents against classical quantitative rules (SMA Crossover, MACD+RSI) and four SOTA LLM trading architectures—*FinMem* [5], *FinAgent* [6], *FinCon* [7], and *TradingAgents* [1]—using a standardized $100,000 initial portfolio over the 2025–2026 out-of-sample window under the identical position-mapping function:

Table 4: Per-Equity Out-of-Sample Net Returns (2025–2026, Frontier Tier).

| Symbol | Mkt | B&H% | M2 Net% | M4 Net% | M4 Turn | $\alpha$ vs M2 |
|---|---|---|---|---|---|---|
| `ASHR` | CN | +21.87 | +17.22 | **+8.38** | 16.0x | −8.84% |
| `600519.SH` | CN | −11.45 | −12.79 | −7.52 | 8.0x | +5.27% |
| `000858.SZ` | CN | −45.23 | −19.91 | −2.88 | 4.0x | +17.04% |
| `300750.SZ` | CN | +4.81 | +13.45 | **+13.64** | 14.0x | +0.19% |
| `002594.SZ` | CN | −32.49 | −25.86 | −7.62 | 8.0x | +18.24% |
| `600036.SH` | CN | +0.95 | −4.36 | −3.66 | 14.0x | +0.70% |
| `601318.SH` | CN | +2.98 | −3.89 | **+19.13** | 8.0x | +23.02% |
| `000333.SZ` | CN | +17.17 | −11.40 | −7.45 | 23.0x | +3.95% |
| `NVDA` | US | +68.88 | +39.42 | **+25.41** | 20.0x | −14.01% |
| `AAPL` | US | +43.79 | +9.18 | −3.29 | 21.0x | −12.47% |
| `GOOGL` | US | +88.05 | +23.17 | **+77.91** | 10.0x | +54.73% |
| `AMZN` | US | +16.83 | +13.22 | −15.88 | 20.0x | −29.09% |

Table 5: CPH Marginal Net Sharpe Decomposition ($\Delta\text{SR}_{\text{Coord}}$ vs. $\Delta\text{SR}_{\text{Scale}}$) and Online Regime Weight Convergence.

| Universe / Metric | $\Delta\text{SR}_{\text{Coord}}$ (SLM) | $\Delta\text{SR}_{\text{Coord}}$ (Frontier) | $\Delta\text{SR}_{\text{Scale}}$ (M4) | Coordination Dominance |
|---|---|---|---|---|
| CN CSI 300 A-Shares | **+0.571** | **+0.833** | -0.097 | Dominates ($\Delta\text{SR}_{\text{Scale}} < 0$) |
| US Tech Benchmark | **+0.915** (Net SR) | **+0.781** (Net SR) | -0.134 | **Turnover Halved** (16.2× **vs** 36.5×) |
| **Final Regime Weights** $w_{i,T}^{(r)}$ | **Market/Tech** | **Fundamentals** | **Macro News** | **Social Sentiment** |
| Bull Regime ($r$ = bull) | **0.342** | 0.276 | 0.238 | 0.144 |
| Bear Regime ($r$ = bear) | 0.216 | **0.227** | **0.336** | **0.221** |
| Sideways ($r$ = sideways) | 0.250 | 0.250 | 0.250 | 0.250 |

- **China CSI 300 A-Shares ($100,000 Initial Capital):** In a challenging down-market where passive Buy-and-Hold loses −$5,410, single-agent baselines lose money (*FinMem*: −$3,830; *FinAgent*: −$2,270), and vanilla multi-agent debate loses −$5,890, **CARE-Agents** delivers **+$2,260 Net PnL** (final equity $102,260, Net Sharpe 0.242, MaxDD 5.89%, $p = 0.0500$), outperforming all other LLM architectures.
- **US Tech Benchmark ($100,000 Initial Capital): CARE-Agents** achieves **+$19,020 Net PnL** (final equity $119,020, Net Sharpe 0.781, turnover 17.75×), maintaining disciplined risk control and cutting portfolio turnover by nearly half

Table 6: SOTA External Baseline PnL ($100,000 Initial Capital) & Risk-Adjusted Comparison on Out-of-Sample 2025–2026 Real Market Data (Frontier Tier, Strictly Symmetric Execution). Compares CARE-Agents against classical quantitative rules (SMA Crossover, MACD+RSI) and published LLM trading architectures: FinMem [5], FinAgent [6], FinCon [7], and TradingAgents [1].

| Universe (Friction) | Method / Published Baseline | Final ($) | Net PnL ($) | Gross% | Net% | Net SR | MaxDD% | Turn. | Brier ↓ | $p$-val |
|---|---|---|---|---|---|---|---|---|---|---|
| CN CSI 300 (15 bps) | Buy & Hold (Market Benchmark) | 94,590 | −5,410 | −5.41 | −5.41 | −0.235 | 13.74 | 0.00 | 0.2897 | 0.2520 |
| CN CSI 300 (15 bps) | SMA Crossover (Classical Quant) | 97,390 | −2,610 | −0.83 | −2.61 | −0.185 | 8.96 | 12.12 | 0.2218 | 0.2460 |
| CN CSI 300 (15 bps) | MACD + RSI Filter (Classical Quant) | 92,210 | −7,790 | −4.92 | −7.79 | −0.845 | 10.79 | 20.38 | 0.2239 | 0.3120 |
| CN CSI 300 (15 bps) | FinMem [5] (Layered Memory) | 96,170 | −3,830 | −0.92 | −3.83 | −0.272 | 10.33 | 19.91 | 0.2249 | 0.1760 |
| CN CSI 300 (15 bps) | FinAgent [6] (KDD'24 Reflection) | 97,730 | −2,270 | +1.39 | −2.27 | −0.136 | 9.30 | 24.53 | 0.2252 | 0.0760 |
| CN CSI 300 (15 bps) | FinCon [7] (NeurIPS'24 CVaR) | 100,050 | +50 | +3.00 | +0.05 | 0.039 | 6.21 | 19.38 | 0.2351 | 0.1040 |
| CN CSI 300 (15 bps) | TradingAgents [1] (Fixed Debate) | 94,110 | −5,890 | −1.57 | −5.89 | −0.591 | 11.25 | 29.93 | 0.2225 | 1.0000 |
| CN CSI 300 (15 bps) | **CARE-Agents (Ours, Full)** | **102,260** | **+2,260** | **+4.09** | **+2.26** | **0.242** | **5.89** | **11.88** | **0.2288** | **0.0500** |
| US Tech (5 bps) | Buy & Hold (Market Benchmark) | 158,440 | +58,440 | +58.44 | +58.44 | 1.210 | 16.72 | 0.00 | 0.3140 | 0.2100 |
| US Tech (5 bps) | SMA Crossover (Classical Quant) | 125,270 | +25,270 | +25.98 | +25.27 | 0.966 | 9.19 | 11.25 | 0.1868 | 0.4600 |
| US Tech (5 bps) | MACD + RSI Filter (Classical Quant) | 106,230 | +6,230 | +7.71 | +6.23 | 0.389 | 13.23 | 27.75 | 0.1920 | 0.2020 |
| US Tech (5 bps) | FinMem [5] (Layered Memory) | 120,470 | +20,470 | +21.73 | +20.47 | 0.805 | 12.24 | 20.80 | 0.2191 | 0.4040 |
| US Tech (5 bps) | FinAgent [6] (KDD'24 Reflection) | 118,910 | +18,910 | +20.31 | +18.91 | 0.729 | 17.37 | 23.43 | 0.2206 | 0.3020 |
| US Tech (5 bps) | FinCon [7] (NeurIPS'24 CVaR) | 124,660 | +24,660 | +25.63 | +24.66 | 1.045 | 8.58 | 15.58 | 0.2311 | 0.3960 |
| US Tech (5 bps) | TradingAgents [1] (Fixed Debate) | 123,260 | +23,260 | +25.29 | +23.26 | 0.926 | 11.36 | 32.86 | 0.2081 | 1.0000 |
| US Tech (5 bps) | **CARE-Agents (Ours, Full)** | **119,020** | **+19,020** | **+20.07** | **+19.02** | **0.781** | **14.92** | **17.75** | **0.2234** | **0.3940** |

compared to *TradingAgents* (17.75× vs. 32.86×).

### 5.4 CPH Marginal Decomposition and Per-Stock Analysis

Table 4 and Table 5 decompose the source of CARE-Agents' outperformance:

- **Empirical Confirmation of CPH:** On China CSI 300 A-Shares, upgrading coordination topology from `M2` to `M4` increases Net Sharpe by +0.571 (SLM) and +0.833 (Frontier). Conversely, scaling the backbone model from 14B SLM to DeepSeek-V3 yields a negative return on `M2` (−0.359) and `M4` (−0.097). Thus, coordination architecture completely dominates backbone scale, validating the CPH.
- **Robust Performance Across Equities:** As shown in Table 4, `M4_CARE_Agents_Ours` outperforms `M2_Vanilla_TradingAgents` on 7 of 8 Chinese equities, achieving massive alpha on trending names (`GOOGL` +54.73% alpha vs. M2; `Ping An` +23.02% alpha vs. M2) while capping drawdown on severe decliners (`000858.SZ` Wuliangye −2.88% vs. −45.23% B&H; `002594.SZ` BYD −7.62% vs. −32.49% B&H).
- **Online Specialist Reweighting:** As shown in the bottom panel of Table 5, `RegimeMultiplicativeWeightManager` dynamically elevates macro `news` (0.336) and `fundamentals` in Bear regimes while maintaining balanced weighting.
- **Live DeepSeek LLM API Verification ($N$ = 15,246 Authentic Traces):** Executing with live DeepSeek-V3 API calls across all 12 assets and 86 weeks (15,246 total cached LLM completions, 0.0% synthetic mock fallback) confirms that `M4` cuts token expenditure from 2,836 to 2,306 tokens/step on CN and from 2,738 to 2,303 tokens/step on US by pruning redundant deliberation rounds via CBS gating.

## 6 Conclusion and R3 Reproducibility Package

We have presented **CARE-Agents**, a mathematically grounded refactoring of multi-agent financial decision pipelines that unifies CBS-gated debate routing, online regime-conditioned Brier multiplicative weights, and counterfactual episodic memory. All source modules, unit test suites, 2025–2026 point-in-time datasets, and benchmark runners (`run_phase5_real_market_benchmark.py`) are packaged for reproduction, satisfying the 2026 R3 Reproducibility standard [3].

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# A Mathematical Proofs

## A.1 Proof of Proposition 1 (Cost-Optimal Debate Gating)

*Proof.* Let $\mathcal{A} = \{1, \dots, K\}$ denote the set of specialists with directional posteriors $p_{i,t} \in \Delta^2$, expected return moves $\hat{m}_{i,t} \geq 0$, and active regime weights $w_t^{(r_t)} \in \Delta^{K-1}$. By Definition 1, the expected pre-reconciliation dispersion value is:

$$\Delta\hat{v}_t = D_{JS}\Big(p_{1,t}, \dots, p_{K,t}; w_t^{(r_t)}\Big) \cdot \sum_{i=1}^{K} w_{i,t}^{(r_t)} \hat{m}_{i,t}, \quad (11)$$

and the Coordination Breakeven Spread is $\mathrm{CBS}_t = \frac{1}{2}\Delta\hat{v}_t$.

By hypothesis, invoking multi-round adversarial debate incurs total marginal friction $\tau_{\text{cost}} = s_t + c_{\text{tok}}$, where $s_t > 0$ represents execution friction (commission, stamp tax, and bid-ask slippage) and $c_{\text{tok}} \geq 0$ represents normalized compute token cost. Moreover, multi-round deliberation reconciles conflicting viewpoints to recover at most a fraction $\gamma \in (0, \frac{1}{2}]$ of the pre-reconciliation dispersion value:

$$\mathbb{E}\left[\Delta\alpha_t\right] \leq \gamma \cdot \Delta\hat{v}_t. \quad (12)$$

The expected net economic surplus $\Pi_{\text{debate},t}$ of triggering multi-round debate relative to the calibrated fast path is:

$$\Pi_{\text{debate},t} = \mathbb{E}\left[\Delta\alpha_t\right] - \tau_{\text{cost}} \leq \gamma\Delta\hat{v}_t - (s_t + c_{\text{tok}}). \quad (13)$$

Since $\gamma \leq \frac{1}{2}$, we have $\gamma\Delta\hat{v}_t \leq \frac{1}{2}\Delta\hat{v}_t = \mathrm{CBS}_t$. Therefore:

$$\Pi_{\text{debate},t} \leq \mathrm{CBS}_t - s_t - c_{\text{tok}} \leq \mathrm{CBS}_t - s_t. \quad (14)$$

If $\mathrm{CBS}_t \leq s_t$, it immediately follows that $\Pi_{\text{debate},t} \leq 0$. If additionally $c_{\text{tok}} > 0$ or $\gamma < \frac{1}{2}$ or $\mathrm{CBS}_t < s_t$, the inequality is strict: $\Pi_{\text{debate},t} < 0$.

Furthermore, if $D_{JS,t} < \tau_{JS}$ where $\tau_{JS} \equiv 2s_t / \sum_{i=1}^{K} w_{i,t}^{(r_t)} \hat{m}_{i,t}$, substitution into Equation (3) gives:

$$\begin{aligned} \mathrm{CBS}_t &= \frac{1}{2} D_{JS,t} \sum_{i=1}^{K} w_{i,t}^{(r_t)} \hat{m}_{i,t} \\ &< \frac{1}{2} \tau_{JS} \sum_{i=1}^{K} w_{i,t}^{(r_t)} \hat{m}_{i,t} = s_t. \end{aligned} \quad (15)$$

Hence, whenever $D_{JS,t} < \tau_{JS}$ or $\mathrm{CBS}_t \leq s_t$, the expected net surplus of invoking debate is strictly negative: $\mathbb{E}[\Pi_{\text{debate},t}] < 0$. Thus, gating debate via $\mathcal{G}_t = \mathbb{I}\{D_{JS,t} \geq \tau_{JS} \wedge \mathrm{CBS}_t > s_t\}$ is cost-optimal. □

## A.2 Proof of Theorem 1 (Regime-Partitioned Brier Regret Bound)

*Proof.* Let $r \in \mathcal{R} = \{\text{bull}, \text{bear}, \text{sideways}\}$ be an arbitrary market regime. Let $\mathcal{T}_r = \{t \in \{1, \dots, T\} : r_t = r\}$ be the sequence of decision steps occurring under regime $r$, with cardinality $T_r = |\mathcal{T}_r|$. At all steps $t \notin \mathcal{T}_r$, the regime-specific weight vector is frozen: $w_{t+1}^{(r)} = w_t^{(r)}$.

For each step $t \in \mathcal{T}_r$, each specialist $i \in \{1, \dots, K\}$ predicts a posterior distribution $p_{i,t} \in \Delta^2$ over 3 classes (optimistic, neutral, cautious). Upon observing true outcome $y_{t+1} \in \{(1,0,0), (0,1,0), (0,0,1)\}$, specialist $i$ incurs multiclass Brier loss:

$$\ell_{i,t} = \frac{1}{3} \sum_{c=1}^{3} \left(p_{i,t}^c - y_{t+1}^c\right)^2. \quad (16)$$

Since $p_{i,t} \in \Delta^2$ and $y_{t+1}$ is a standard basis vector, the maximum Euclidean distance on the probability simplex satisfies $\sum_{c=1}^{3}(p_{i,t}^c - y_{t+1}^c)^2 \leq 2$. Consequently, $\ell_{i,t}$ is strictly bounded in $[0, L]$ where $L = \frac{2}{3}$.

The ensemble loss at step $t \in \mathcal{T}_r$ under weight vector $w_t^{(r)}$ is $\ell_{\mathrm{MWU},t} = \sum_{i=1}^K w_{i,t}^{(r)} \ell_{i,t}$. Specialists are updated via the standard Exponential Forecaster:

$$w_{i,t+1}^{(r)} = \frac{w_{i,t}^{(r)} \exp(-\eta_r \ell_{i,t})}{\sum_{j=1}^K w_{j,t}^{(r)} \exp(-\eta_r \ell_{j,t})}, \tag{17}$$

with uniform initial weights $w_{i,1}^{(r)} = 1/K$. Define the potential function $W_t = \sum_{i=1}^K w_{i,t}^{(r)} = 1$ and normalizer $Z_t = \sum_{i=1}^K w_{i,t}^{(r)} \exp(-\eta_r \ell_{i,t})$.

By Hoeffding's Lemma, for any random variable $X \in [0, L]$ and scalar $\lambda \in \mathbb{R}$, $\mathbb{E}[e^{-\lambda X}] \le \exp(-\lambda \mathbb{E}[X] + \frac{\lambda^2 L^2}{8})$. Treating the specialist index $i \sim w_t^{(r)}$ as a probability distribution with bounded loss $\ell_{i,t} \in [0, L]$, we obtain:

$$\begin{aligned} Z_t &= \sum_{i=1}^K w_{i,t}^{(r)} \exp(-\eta_r \ell_{i,t}) \\ &\le \exp\left( -\eta_r \sum_{i=1}^K w_{i,t}^{(r)} \ell_{i,t} + \frac{\eta_r^2 L^2}{8} \right) \\ &= \exp\left( -\eta_r \ell_{\mathrm{MWU},t} + \frac{\eta_r^2 L^2}{8} \right). \end{aligned} \tag{18}$$

Taking logarithms and summing across all $t \in \mathcal{T}_r$:

$$\sum_{t \in \mathcal{T}_r} \ln Z_t \le -\eta_r \sum_{t \in \mathcal{T}_r} \ell_{\mathrm{MWU},t} + \frac{\eta_r^2 L^2 T_r}{8}. \tag{19}$$

Conversely, by definition of the telescoping weight update, for any fixed specialist $k \in \{1, \dots, K\}$:

$$\begin{aligned} \sum_{t \in \mathcal{T}_r} \ln Z_t &= \ln \left( \sum_{i=1}^K w_{i,1}^{(r)} \exp\left( -\eta_r \sum_{t \in \mathcal{T}_r} \ell_{i,t} \right) \right) \\ &= \ln \left( \frac{1}{K} \sum_{i=1}^K \exp\left( -\eta_r \sum_{t \in \mathcal{T}_r} \ell_{i,t} \right) \right) \\ &\ge \ln \left( \frac{1}{K} \exp\left( -\eta_r \sum_{t \in \mathcal{T}_r} \ell_{k,t} \right) \right) \\ &= -\ln K - \eta_r \sum_{t \in \mathcal{T}_r} \ell_{k,t}. \end{aligned} \tag{20}$$

Combining Equation (19) and Equation (20):

$$-\ln K - \eta_r \sum_{t \in \mathcal{T}_r} \ell_{k,t} \le -\eta_r \sum_{t \in \mathcal{T}_r} \ell_{\mathrm{MWU},t} + \frac{\eta_r^2 L^2 T_r}{8}. \tag{21}$$

Rearranging terms and dividing by $\eta_r > 0$:

$$\sum_{t \in \mathcal{T}_r} \ell_{\mathrm{MWU},t} - \sum_{t \in \mathcal{T}_r} \ell_{k,t} \le \frac{\ln K}{\eta_r} + \frac{\eta_r L^2 T_r}{8}. \tag{22}$$

Because this inequality holds for every specialist $k \in \{1, \dots, K\}$, it holds for the empirical best specialist $i_r^* = \arg\min_{i \in \mathcal{A}} \sum_{t \in \mathcal{T}_r} \ell_{i,t}$:

$$\begin{aligned} \mathrm{Regret}_r(T_r) &= \sum_{t \in \mathcal{T}_r} \ell_{\mathrm{MWU},t} - \min_{i \in \mathcal{A}} \sum_{t \in \mathcal{T}_r} \ell_{i,t} \\ &\le \frac{\ln K}{\eta_r} + \frac{\eta_r L^2 T_r}{8}. \end{aligned} \tag{23}$$

Minimizing this convex upper bound with respect to $\eta_r$ yields the optimal learning rate $\eta_r^* = \sqrt{8 \ln K / (L^2 T_r)}$. Substituting $\eta_r^*$ and $L = \frac{2}{3}$:

$$\begin{aligned} \mathrm{Regret}_r(T_r) &\le 2\sqrt{\frac{\ln K \cdot L^2 T_r}{8}} \\ &= L\sqrt{\frac{T_r \ln K}{2}} = \frac{2}{3}\sqrt{\frac{T_r \ln K}{2}}. \end{aligned} \tag{24}$$

This completes the proof of Theorem 1. □

# B Live DeepSeek API Verification and Posterior Audit

To provide complete empirical verification on live frontier LLM inference across our full evaluation horizon, we executed `scripts/run_phase5_real_market_benchmark.py --use-live-llm-api` against the production DeepSeek API (`https://api.deepseek.com/v1/chat/completions`) across all 12 equities (8 China CSI 300 A-Shares and 4 US Tech equities) over all 86 out-of-sample decision weeks (12 assets × 86 weeks = 1,032 asset-weeks across specialists, multi-round debate/roleplay protocols, and SOTA baseline agents, totaling 15,246 live LLM completions archived with full prompts, raw completions, timestamps, and measured `usage.total_tokens` in `results/live_llm_posteriors_cache.json`).

Table 7: Live DeepSeek API Verification (15,246 Authentic Traces, Zero Mock Fallback) and Protocol Comparison.

| Live Specialist / Protocol | Calls | $\bar{p}_{\mathrm{opt}}$ | $\bar{p}_{\mathrm{neu}}$ | $\bar{p}_{\mathrm{cau}}$ | Brier ↓ | Avg Tok |
|---|---|---|---|---|---|---|
| `market` Specialist (Live API) | 1,032 | 0.312 | 0.348 | 0.340 | — | 343.8 |
| `fundamentals` Specialist (Live API) | 1,032 | 0.316 | 0.354 | 0.330 | — | 347.1 |
| `news` Specialist (Live API) | 1,032 | 0.310 | 0.366 | 0.324 | — | 344.5 |
| `social` Specialist (Live API) | 1,032 | 0.311 | 0.351 | 0.338 | — | 341.9 |
| CN Benchmark: `M2_Vanilla_TradingAgents` | 688 | — | — | — | 0.2225 | 2,836 |
| CN Benchmark: `M4_CARE_Agents_Ours` | **688** | — | — | — | **0.2288** | **2,306** |
| US Benchmark: `M2_Vanilla_TradingAgents` | 344 | — | — | — | 0.2081 | 2,738 |
| US Benchmark: `M4_CARE_Agents_Ours` | **344** | — | — | — | **0.2234** | **2,303** |

**Verbatim Cached Live DeepSeek Trace (`600519.SH`, `2025-02-28`, 345 tokens):** *"Kweichow Moutai shows a constructive short-term setup with a 4.51% three-week return, price trading 2.95% above its 8-day MA, and a bull causal regime, while a 21.4x PE and 3.73% dividend yield provide valuation support at current levels...* `{"p_optimistic": 0.45, "p_neutral": 0.35, "p_cautious": 0.20, "expected_move_bps": 60.0, "confidence": 0.62, "market_regime": "bull"}`*".*